

Basketball Live Trading

Signal Design & Decision Rules

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Executive Summary

This document defines the signal design, filtering logic, and execution rules for live basketball trading. The central insight is that basketball's continuous, high-frequency scoring makes raw score differentials almost meaningless as trading signals. A single point — or even a 4-point swing — says almost nothing about win probability in most game states.

The framework therefore replaces raw score logic with game-state-normalised signals, layered through a regime gate, persistence filter, and edge threshold before any execution is triggered.

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Never trade the score. Trade the edge — only when it is stable, normalised, and survives all filters.

1. The Basketball Volatility Problem

Basketball is unique among major sports in several ways that directly complicate live trading signals:

- Scoring is near-continuous. A 4-point swing takes under 30 seconds and carries minimal win-probability information in most game states.
- Foul accumulation creates lottery-like sequences. Three fouls in two minutes flips the game into a free-throw-dominated state your model has not been trained on.
- End-of-quarter dynamics are structurally different. Intentional fouling, timeout stacking, and clock manipulation create a regime that is not representative of normal play.
- Early leads are predictively weak. A 10-point Q1 lead is historically unreliable as a win predictor. Your model may handle this correctly — but a regime filter should enforce it regardless.

The core consequence: a model update every second is not a trading signal every second. The signal pipeline must be designed to suppress the noise inherent to basketball before a trade decision is made.

2. The Three-Layer Architecture

Every trade decision passes through three distinct layers. These are not optional — each exists to filter a different class of noise.

Layer	Purpose	Filters out
Signal	Model + market producing an edge reading	Nothing yet — raw input
Decision rule	Is signal strong and stable enough to act?	Noise, flickers, bad regimes
Execution	Size, limit vs market, venue	Only reached after rule passes

Your current loop generates a signal every second. The decision rule layer is what prevents that from becoming a trade every second.

3. Normalise by Game State, Not Raw Score

A raw score difference is almost never the right input. What matters is the lead relative to expected variance at the current game state.

3.1 Normalised Lead

The correct signal input is:

```
normalised_lead = score_diff /  $\sigma$ (score_diff | quarter, time_remaining)
```

σ here is empirical — it captures how much the score typically swings from this point in the game. It is not a fixed number. It varies substantially:

Quarter	Time remaining	Typical σ (pts)	Implication
Q1	10 min	~14–18	Even a 10pt lead is within 1σ
Q2	5 min	~10–14	Score matters more, not decisive
Q4	3 min	~6–9	Lead starts to matter significantly
Q4	45 sec	~2–4	Game state nearly resolved

Only when `normalised_lead` moves meaningfully does the signal deserve attention. Do not pass raw `score_diff` into your decision logic.

3.2 Scale the Edge Threshold by Time

A 3-cent edge in Q2 is worth less than a 3-cent edge in Q4 with 8 minutes remaining. The threshold should reflect how much time remains for mean reversion:

$$\epsilon_{\text{effective}} = \epsilon_{\text{base}} \times (1 + k \times \text{time_remaining_fraction})$$

This means you require a larger edge early in the game — when there is more time for the market to close against you — and allow a tighter edge late, when the game state is more resolved.

4. Regime Gate — Dead Zones

These are game states where the volatility is structurally untradeable. The regime gate sits before any edge check. If any condition fires, the system moves to IDLE immediately.

4.1 Dead Zone Definitions

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3+ fouls in 2 minutes. Win probability can swing 5–8% on outcomes that are ~73% coin-flip. Do not trade during or for 90 seconds after the cluster ends.

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Last 60 seconds of any quarter. Intentional fouling, timeout stacking, clock manipulation. Market makers widen spreads here for a reason.

EARLY BLOWOUT

Q1 lead ≥ 10 points, or Q2 lead ≥ 18 points. Historically weak predictors. Your model may correctly flag low edge; the regime gate enforces it unconditionally.

TIMEOUT CLUSTER

Back-to-back timeouts within 90 seconds. Coach adjustments reset momentum in ways the live model cannot observe.

4.2 Re-entry Condition

After a dead zone clears, do not immediately re-enter signal evaluation. Require:

- 2 full minutes of clean play (no fouls, no timeouts) since the dead zone ended, AND
- Edge has persisted above ϵ for the full K-tick window (no partial credit for stability before the dead zone).

5. Persistence Filter

Because scoring in basketball is so continuous, a single bucket or turnover legitimately flickers the model. A snapshot edge is not a trading signal.

5.1 Requirements

- Same directional side for K consecutive ticks (recommended: 30–60 seconds)
- EMA of edge, not the raw snapshot reading
- K should be longer than in most other sports given basketball's scoring frequency

5.2 Suggested Parameters

Parameter	Recommended value	Notes
K (persistence ticks)	30–60 seconds	Start at 45s; tune via backtest
EMA span	20–30 ticks	Smooth single-play flickers
Side consistency	100% same side in window	Any flip resets the counter

6. Edge Threshold

Only consider execution when the smoothed edge clears the threshold after spread and fees.

6.1 Minimum Edge

- Start with ϵ_{base} = 3–5 cents in probability space, net of all transaction costs
- Apply time-scaling (Section 3.2) to make the threshold dynamic
- Consider Δedge rather than level when the market is actively drifting: trade when edge increases by Δ from your last decision baseline, not just when edge is positive

6.2 Re-entry Threshold (Hysteresis)

After any trade or near-trade, the re-entry bar is higher:

$$\text{re-entry threshold} = \epsilon \times 1.5$$

This prevents thrashing around the threshold boundary, which is a common failure mode when edge is borderline.

7. State Machine

Treat trading as a state machine with explicit transitions. Every transition should be logged with the reason it fired.

State	Entry condition	Exit to	Exit condition
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IDLE	Default; also after regime gate fires or cooldown	WATCHING	Clean regime + edge begins building
WATCHING	Edge $> \epsilon$ first appears on EMA	ARMED or IDLE	ARMED if K-tick persistence met; IDLE if edge flips
ARMED	Persistence met; awaiting final check	EXECUTE or IDLE	EXECUTE if edge still $> \epsilon$; IDLE on flip
EXECUTE	All conditions passed	COOLDOWN	Immediately post-execution
COOLDOWN	Post-execution lockout	IDLE	M minutes elapsed; re-entry at $\epsilon \times 1.5$

8. Horizon Clarity: Snapshot vs. 1-Minute Ahead

There are two honest interpretations of trading with a 1-minute horizon. Be explicit about which you are implementing.

Option A: Predict market price 1 minute ahead

This is a different modelling target than $P(\text{home wins})$ now. You would predict the market mid-price at $t+60s$ using current features plus autoregressive market terms. Your current model does not output this unless trained for it. This is a separate modelling project.

Option B: Sample the snapshot more slowly (recommended starting point)

Simpler and correct as a starting point. Every ~60 seconds, recompute snapshot $\rightarrow$ model $\rightarrow$ edge; apply all filters above. You are not forecasting the market 1 minute ahead; you are sampling slower and filtering aggressively. Most systematic live traders begin here.

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Start with Option B. Option A requires labelled forward market price targets — a separate data engineering project. Do not conflate the two.

9. Cost Rationality

Every trade pays spread + fees + adverse selection. This forces a simple inequality check before execution:

```
expected edge × size > spread + fees + variance_penalty
```

In practice this means:

- Smaller frequency and larger ϵ is often more rational than trading more often at thinner edges
- Adverse selection risk is higher during volatile game states — another reason the regime gate matters
- Paper-backtest your cooldown and ϵ combination before going live; the cost-adjusted edge will look very different from the gross edge

10. Implementation Checklist

In order of implementation priority:

#	Task	Priority
1	Replace raw score_diff with normalised_lead (score_diff / σ)	Critical
2	Implement regime gate: foul storm, end-of-quarter, early blowout, timeout cluster	Critical
3	Add EMA smoothing to edge signal (20–30 tick span)	High
4	Implement K-tick persistence counter (reset on side flip)	High
5	Implement state machine: IDLE → WATCHING → ARMED → EXECUTE → COOLDOWN	High
6	Apply time-scaled ϵ threshold ($\epsilon_{\text{base}} \times \text{time multiplier}$)	Medium
7	Log every state transition with timestamp and reason	Medium

8	Paper-backtest with cooldown and ϵ combinations	Medium
9	Tune K and EMA span against backtest P&L	Low
10	Consider Option A (forward market price model) only after Option B is stable	Future